Beyond one shot: agentic LaTeX → Typst conversion

Four one-factor upgrades, ablated on the 6 hardest lathe benchmark samples

37 → 82

mean overall fidelity, 1-shot baseline vs best method (core 6)

$77

total tracked API spend

72

runs completed

2026-07-14 - harness_baseline/ - metric pdf_fidelity_v0.1 (deterministic)

Setup - task, metric, and gates

Task set

The 6 hardest of 157 benchmark samples (one-shot LLM overall 31-38/100): four table documents that reflow to the wrong page count, one algorithm block, one figure+caption page. Reference = pdfLaTeX render.

Pass gates (frozen before running)

G1 compiles - G2 page count and pagination perfect - G3 token precision AND recall >= 0.95 - G4 layout >= 75.

Score (deterministic, 0-100)

overall = content^0.35 x visual^0.65; visual = pagination^0.10 x layout^0.40 x typography^0.20 x raster^0.30. Raster uses the raster_v0.2 rescoring (tolerant ink F1 + edge distance) which removes v0.1's ~25-30 harshness ceiling; identity still scores 100 and orderings are preserved.

How to read this deck

Each following section changes exactly ONE factor (prompt->harness, effort, visual feedback, tooling) and shows the same table: per-metric means over the core 6, cost, duration, iterations - then a case study with the agent's own reasoning per scoring iteration.

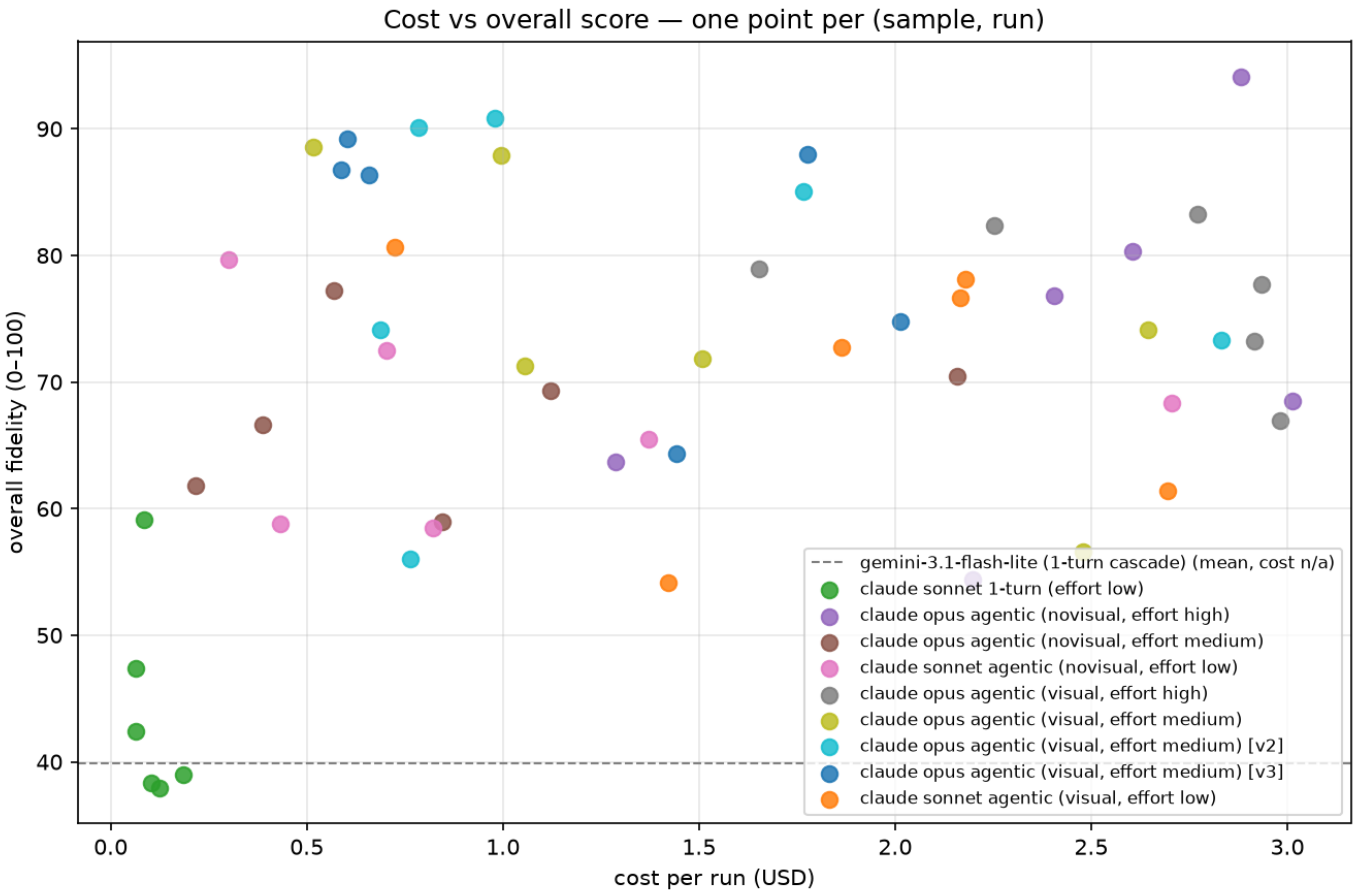
Leaderboard - core 6 hardest samples

all metrics 0-100; mean over samples

model	effort	feedback	harness	overall	content	layout	typo	raster	pagin.	pass	$/run
opus-4.7	medium	visual	v3	**81.6**	95.0	80.6	83.5	58.9	100.0	2/6	1.18
opus-4.7	medium	visual	v2	**78.2**	96.4	77.2	83.7	51.7	99.1	3/6	1.30
opus-4.7	high	visual	v1	**77.1**	97.0	80.0	84.9	42.7	100.0	4/6	2.58
opus-4.7	medium	visual	v1	**75.1**	96.6	71.4	81.5	46.8	99.7	2/6	1.53
opus-4.7	high	blind	v1	**73.0**	96.5	79.4	87.2	36.1	99.0	3/6	2.40
sonnet-4.6	low	visual	v1	**70.6**	95.6	67.7	78.7	38.0	100.0	0/6	1.84
opus-4.7	low	visual	v1	**(69.9)**	(95.3)	(71.7)	(77.8)	(33.5)	(100.0)	1/2	-
opus-4.7	medium	blind	v1	**67.4**	94.7	70.5	81.5	28.6	100.0	0/6	0.88
sonnet-4.6	low	blind	v1	**67.2**	92.1	69.3	85.6	29.6	100.0	0/6	1.06
opus-4.7	low	blind	v1	**66.8**	94.1	65.0	84.6	32.7	92.5	2/6	-
sonnet-4.6	low	1-turn	—	**(43.4)**	(90.1)	(34.4)	(76.2)	(10.4)	(65.4)	0/6	0.10
opus-4.7	low	1-turn	—	**(41.5)**	(89.4)	(34.0)	(71.4)	(9.6)	(60.0)	0/6	-
gemini-3.1-flash-lite	—	1-turn	—	**37.0**	83.5	30.6	76.1	7.3	56.9	0/6	-

Parenthesized = partial coverage (queues still running). The four highlighted one-factor steps from the bottom of this table to the top are ablated in the following sections.

Cost vs score



Reading the plot:

- One-shot points (bottom-left, dashed gemini line) cluster at 32-52 regardless of model.

- Best value sits at $0.5-1.0 per document.

- Spending to the $3 cap does not buy score: those points are budget-cap grinders oscillating around a plateau.

- Implication: feedback loop >> model scale >> raw budget.

Upgrade 1: one-shot prompt -> agentic harness

What changed in this step

Same model (opus-4.7), same low effort, still no images. The only change: instead of one prompt -> one answer, the model gets a workdir with main.tex, ./score.sh (compile + official metrics), and freedom to edit-score-edit under a $3 cap. Feedback is numeric only (blind).

	overall	content	layout	typo	raster	pagin.	pass	$/run	min/run	iters
BEFORE opus-4.7 / low / 1-turn	(41.5)	(89.4)	(34.0)	(71.4)	(9.6)	(60.0)	0/6	-	-	-
AFTER opus-4.7 / low / blind / v1	66.8	94.1	65.0	84.6	32.7	92.5	2/6	-	-	-
Δ	**+25.3**	**+4.7**	**+31.0**	**+13.2**	**+23.1**	**+32.5**	**+2**	-	-	-

Per-sample overall (before -> after):

06_tables_moderate_010: 39.0 -> 76.9 (+38.0)
05_tables_simple_023: 35.4 -> 68.4 (+33.0)
05_tables_simple_005: 38.0 -> 82.9 (+44.9)
09_algorithms_003: 56.5 -> 46.6 (-9.9)
05_tables_simple_021: 38.6 -> 43.9 (+5.3)
07_figures_captions_007: nc/- -> 81.9

Upgrade 1: one-shot prompt -> agentic harness - case: 05_tables_simple_005

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / low / 1-turn	38.0	86.9	24.6	65.1	9.6	53.5
AFTER opus-4.7 / low / blind / v1	82.9	99.9	78.5	87.5	57.8	100.0

No transcript for the AFTER run (older run without step logging).

Upgrade 2: reasoning effort low -> medium

What changed in this step

Identical harness, identical blind feedback. Only the CLI --effort flag changes, giving the model a larger thinking budget per turn.

	overall	content	layout	typo	raster	pagin.	pass	$/run	min/run	iters
BEFORE opus-4.7 / low / blind / v1	66.8	94.1	65.0	84.6	32.7	92.5	2/6	-	-	-
AFTER opus-4.7 / medium / blind / v1	67.4	94.7	70.5	81.5	28.6	100.0	0/6	0.88	2.8	8
Δ	**+0.6**	**+0.6**	**+5.5**	**-3.2**	**-4.1**	**+7.5**	**-2**	-	-	-

Per-sample overall (before -> after):

06_tables_moderate_010: 76.9 -> 69.3 (-7.6)
05_tables_simple_023: 68.4 -> 70.5 (+2.1)
05_tables_simple_005: 82.9 -> 77.2 (-5.6)
09_algorithms_003: 46.6 -> 61.8 (+15.2)
05_tables_simple_021: 43.9 -> 59.0 (+15.0)
07_figures_captions_007: 81.9 -> 66.6 (-15.3)

Upgrade 2: reasoning effort low -> medium - case: 09_algorithms_003

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / low / blind / v1	46.6	94.1	36.7	85.3	10.1	79.9
AFTER opus-4.7 / medium / blind / v1	61.8	94.9	71.2	80.8	16.9	100.0

AFTER run, iteration by iteration (opus-4.7 / medium / blind / v1):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	55.1	start	write: “#set page(paper: "us-letter", margin: 1in) #set text(size: 11pt, font: "New Comp”

Upgrade 3: blind -> visual feedback

What changed in this step

Same model, effort, harness. The agent may now Read reference page PNGs and the labeled diff images (red = missing, blue = extra, yellow/orange = displaced) regenerated by every score run.

	overall	content	layout	typo	raster	pagin.	pass	$/run	min/run	iters
BEFORE opus-4.7 / medium / blind / v1	67.4	94.7	70.5	81.5	28.6	100.0	0/6	0.88	2.8	8
AFTER opus-4.7 / medium / visual / v1	75.1	96.6	71.4	81.5	46.8	99.7	2/6	1.53	4.4	10
Δ	**+7.6**	**+1.9**	**+1.0**	**-0.0**	**+18.2**	**-0.3**	**+2**	**+0.65**	-	-

Per-sample overall (before -> after):

06_tables_moderate_010: 69.3 -> 74.1 (+4.8)
05_tables_simple_023: 70.5 -> 56.6 (-13.9)
05_tables_simple_005: 77.2 -> 71.3 (-6.0)
09_algorithms_003: 61.8 -> 87.9 (+26.1)
05_tables_simple_021: 59.0 -> 71.9 (+12.9)
07_figures_captions_007: 66.6 -> 88.5 (+21.9)

Upgrade 3: blind -> visual feedback - case: 09_algorithms_003

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / medium / blind / v1	61.8	94.9	71.2	80.8	16.9	100.0
AFTER opus-4.7 / medium / visual / v1	87.9	96.8	92.0	85.4	67.9	100.0

AFTER run, iteration by iteration (opus-4.7 / medium / visual / v1):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	61.2	start	edit: “#ind(2)[*if* $P_(u,t) > #text(font: "DejaVu Sans Mono", size: 10pt)[MinimumPower”
#02	61.2	+0.0	edit: “#set par(justify: true, leading: 0.55em, first-line-indent: 0pt)”
#03	72.9	+11.7	edit: “#set par(justify: false, leading: 0.45em, spacing: 0.5em) #show math.equation: s”
#04	74.0	+1.1	edit: “#set par(justify: false, leading: 0.5em, spacing: 0.55em) #show math.equation: s”
#05	78.2	+4.2	edit: “#set par(justify: false, leading: 0.55em, spacing: 0.7em) #show math.equation: s”
#06	74.6	-3.6	edit: “#v(-0.5cm) #line(length: 100%, stroke: 0.8pt) #v(-0.5em) #line(length: 100%, str”
#07	78.2	+3.6	edit: “#line(length: 100%, stroke: 0.8pt) #v(-0.4em) #line(length: 100%, stroke: 0.4pt)”

Upgrade 4: harness v1 -> v2 tooling

What changed in this step

Same model, effort, visual feedback. v2 adds: best-so-far auto-checkpoint (regressions become free); ./sweep.sh deterministic grid search over typst --input values; fine-grained per-component AND per-page score report; registered-raster diagnostic (separates misalignment from the font floor).

	overall	content	layout	typo	raster	pagin.	pass	$/run	min/run	iters
BEFORE opus-4.7 / medium / visual / v1	75.1	96.6	71.4	81.5	46.8	99.7	2/6	1.53	4.4	10
AFTER opus-4.7 / medium / visual / v2	78.2	96.4	77.2	83.7	51.7	99.1	3/6	1.30	4.2	8
Δ	**+3.2**	**-0.2**	**+5.8**	**+2.2**	**+4.9**	**-0.5**	**+1**	**-0.23**	-	-

Per-sample overall (before -> after):

06_tables_moderate_010: 74.1 -> 74.1 (-0.0)
05_tables_simple_023: 56.6 -> 73.3 (+16.7)
05_tables_simple_005: 71.3 -> 90.8 (+19.6)
09_algorithms_003: 87.9 -> 85.0 (-2.9)
05_tables_simple_021: 71.9 -> 56.1 (-15.8)
07_figures_captions_007: 88.5 -> 90.1 (+1.6)

Upgrade 4: harness v1 -> v2 tooling - case: 05_tables_simple_005

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / medium / visual / v1	71.3	99.9	58.2	90.1	38.9	100.0
AFTER opus-4.7 / medium / visual / v2	90.8	99.4	87.7	89.9	78.9	100.0

AFTER run, iteration by iteration (opus-4.7 / medium / visual / v2):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	68.4	start	write: “#set page(paper: "us-letter", margin: 1in) #set text(font: "New Computer Modern"”
#02	65.5	-2.9	Content is 99. Table rows are taller than reference and shifted down. Let me tighten row spacing. edit: “ inset: (x: 4pt, y: 1.2pt), table.header(text(size: 8pt)[*Index*], text(size: 8”
#03	68.6	+3.1	edit: “ inset: (x: 4pt, y: 1.5pt), table.header(text(size: 7pt)[*Index*], text(size: 7”
#04	68.6	+0.0	edit: “#let topmargin = eval(sys.inputs.at("topmargin", default: "1in")) #set page(pape”
#05	76.4	+7.8	0 edit(s) before scoring
#06	76.4	+0.0	0 edit(s) before scoring
#07	80.7	+4.3	edit: “#pagebreak() #v(p2gap) #make_table[Table 2: Source table 2: 2512.00323_table_10]”
#08	82.0	+1.3	0 edit(s) before scoring
#09	82.0	+0.0	edit: “#let p2gap = eval(sys.inputs.at("p2gap", default: "70pt"))”

Upgrade 5: harness v2 -> v3 (fix the mid-run reward)

What changed in this step

Same model, effort, visual feedback, and v2 tooling. One change: the score the agent sees MID-RUN now uses raster_v0.2 (tolerant ink F1 + edge distance) - the same number as final grading - instead of v0.1's harsh raster whose ~25-30 ceiling made real improvements invisible mid-run. The agent stops wasting budget chasing a dead component and optimizes what actually moves.

	overall	content	layout	typo	raster	pagin.	pass	$/run	min/run	iters
BEFORE opus-4.7 / medium / visual / v2	78.2	96.4	77.2	83.7	51.7	99.1	3/6	1.30	4.2	8
AFTER opus-4.7 / medium / visual / v3	81.6	95.0	80.6	83.5	58.9	100.0	2/6	1.18	3.6	9
Δ	**+3.3**	**-1.5**	**+3.4**	**-0.1**	**+7.2**	**+0.9**	**-1**	**-0.12**	-	-

Per-sample overall (before -> after):

06_tables_moderate_010: 74.1 -> 86.4 (+12.3)
05_tables_simple_023: 73.3 -> 64.4 (-8.9)
05_tables_simple_005: 90.8 -> 88.0 (-2.8)
09_algorithms_003: 85.0 -> 86.8 (+1.7)
05_tables_simple_021: 56.1 -> 74.8 (+18.7)
07_figures_captions_007: 90.1 -> 89.2 (-0.9)

Upgrade 5: harness v2 -> v3 (fix the mid-run reward) - case: 05_tables_simple_021

	overall	content	layout	typo	raster	pagin.
BEFORE opus-4.7 / medium / visual / v2	56.1	90.4	42.9	65.5	25.7	94.7
AFTER opus-4.7 / medium / visual / v3	74.8	88.8	71.2	77.5	52.0	99.9

AFTER run, iteration by iteration (opus-4.7 / medium / visual / v3):

iter	overall	Δ	agent's reasoning before this score run / last edit
#01	53.8	start	write: “#set page(paper: "us-letter", margin: 1in) #set text(font: "New Computer Modern"”
#02	61.0	+7.2	Table 2 needs to be more compact. Let me reduce font size and row spacing. edit: “// ============ TABLE 2 ============ #set text(size: 8pt) #set par(leading: 0.4e”
#03	63.1	+2.1	edit: “// ============ TABLE 2 ============ #set text(size: 8pt) #set par(leading: 0.4e”

Best conversion so far: 05_tables_simple_005

opus-4.7 / high / blind / v1 - overall 85.3

Reference (pdfLaTeX)

Simple Tables

Source-backed grouped table sample

1 Tables

These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.00323_table_8

Index	Model Name	Model Size (MB)	Dataset
1	twmkn9/albert-base-v2-squad2	12	SQuAD v2
2	valhalla/bart-large-finetuned-squadv1	388	SQuAD v1
3	deepset/bert-base-cased-squad2	104	SQuAD v2
4	google/bigbird-roberta-base	122	Books, CC-News, Stories and Wikipedia
5	google/bigbird-pegasus-large-arxiv	551	Arxiv dataset
6	dmis-lab/biobert-v1.1	104	NA
7	deepset/roberta-base-squad2	129	SQuAD v2
8	Splend1dchan/canine-c-squad	126	NA
9	YituTech/conv-bert-base	101	NA
10	Palak/microsoft_deberta-large_squad	387	SQuAD v1
11	microsoft/deberta-v2-xlarge	844	NA
12	distilbert-base-uncased	64	BookCorpus and English Wikipedia
13	bhadresh-savani/electra-base-squad2	104	SQuAD v2
14	nghuyong/ernie-1.0-base-zh	96	Chinese
15	xlm-mlm-en-2048	637	Masked language modeling
16	google/bert-base	80	Colossal Clean Crawled Corpus (C4)
17	funnel-transformer/small	125	BookCorpus, English Wikipedia, Clue Web, GigaWord, and Common Crawl
18	EleutherAI/gpt-neo-1.3B	1255	Pile
19	hf-internal-testing/tiny-random-gpt j	1	NA
20	gpt2	119	WebText
21	lysteven/ibert-roberta-base	119	NA
22	allenai/led-base-16384	155	NA
23	allenai/longformer-large-4096-finetuned-triviaqa	415	NA
24	facebook/mbart-large-cc25	583	Multilingual mbart model
25	sanayier/nezha-base-wikitext	8	wikitext-103
26	csarron/mobilebert-uncased-squad-v2	24	SQuAD v2
27	microsoft/mpnet-base	105	NA
28	google/mt5-small	165	101 languages
29	RUCAIBox/mvp	388	NA
30	ajitabc/nezha-cn-base	98	Chinese
31	uw-madison/nystromformer-512	104	BookCorpus and English Wikipedia
32	facebook/opt-350m	316	BookCorpus, CC-Stories, The Pile, Pushshift.io, and CCNewsV2
33	bert-base-uncased	105	BookCorpus and English Wikipedia
34	google/rembert	550	Multilingual Wikipedia data over 110 languages.
35	roberta-base	119	BookCorpus, English Wikipedia, CC-News, OpenWebText, and Stories
36	andreasmadsen/efficient_mlm_m0.40	339	NA
37	ArthurZ/dummy-rochert-qa	112	NA
38	tau/splinter-base	103	BookCorpus and English Wikipedia
39	squeezebert/squeezebert-uncased	49	BookCorpus and English Wikipedia
40	t5-small	58	Colossal Clean Crawled Corpus (C4)
41	xlm-mlm-en-2048	637	Masked language modeling
42	xlnet-base-cased	112	XLNet model pre-trained on English language
43	uw-madison/yoso-4096	121	Masked language modeling
44	SRDdev/QABERT-small	64	30k samples from the Stanford Question Answering Dataset
45	bert-large-uncased-whole-word-masking-finetuned-squad	320	BookCorpus and English Wikipedia
46	facebook/bart-large-cnn	388	CNN Daily Mail
47	ahotrod/electra_large_discriminator_squad2_512	319	SQuAD v2

1

Final candidate (Typst)

Simple Tables

Source-backed grouped table sample

1 Tables

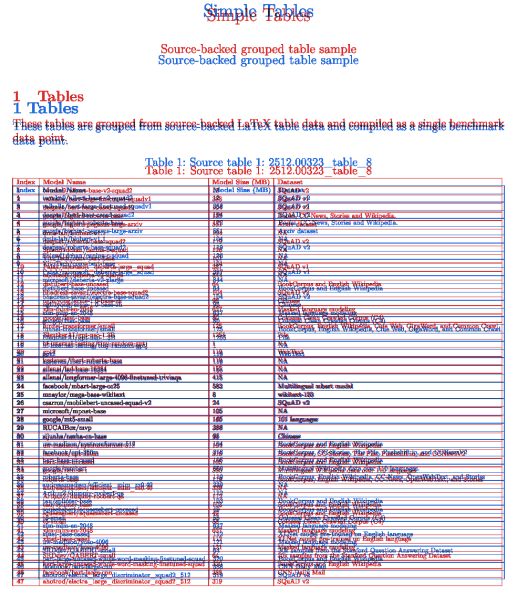
These tables are grouped from source-backed LaTeX table data and compiled as a single benchmark data point.

Table 1: Source table 1: 2512.00323_table_8

Index	Model Name	Model Size (MB)	Dataset
1	twmkn9/albert-base-v2-squad2	12	SQuAD v2
2	valhalla/bart-large-finetuned-squadv1	388	SQuAD v1
3	deepset/bert-base-cased-squad2	104	SQuAD v2
4	google/bigbird-roberta-base	122	Books, CC-News, Stories and Wikipedia.
5	google/bigbird-pegasus-large-arxiv	551	Arxiv dataset
6	dmis-lab/biobert-v1.1	104	NA
7	deepset/roberta-base-squad2	119	SQuAD v2
8	Splend1dchan/canine-c-squad	126	NA
9	YituTech/conv-bert-base	101	NA
10	Palak/microsoft_deberta-large_squad	387	SQuAD v1
11	microsoft/deberta-v2-xlarge	844	NA
12	distilbert-base-uncased	64	BookCorpus and English Wikipedia
13	bhadresh-savani/electra-base-squad2	104	SQuAD v2
14	nghuyong/ernie-1.0-base-zh	96	Chinese
15	xlm-mlm-en-2048	637	Masked language modeling
16	google/bert-base	80	Colossal Clean Crawled Corpus (C4)
17	funnel-transformer/small	125	BookCorpus, English Wikipedia, Clue Web, GigaWord, and Common Crawl
18	EleutherAI/gpt-neo-1.3B	1255	Pile
19	hf-internal-testing/tiny-random-gpt j	1	NA
20	gpt2	119	WebText
21	lysteven/ibert-roberta-base	119	NA
22	allenai/led-base-16384	155	NA
23	allenai/longformer-large-4096-finetuned-triviaqa	415	NA
24	facebook/mbart-large-cc25	583	Multilingual mbart model
25	sanayier/nezha-base-wikitext	8	wikitext-103
26	csarron/mobilebert-uncased-squad-v2	24	SQuAD v2
27	microsoft/mpnet-base	105	NA
28	google/mt5-small	165	101 languages
29	RUCAIBox/mvp	388	NA
30	ajitabc/nezha-cn-base	98	Chinese
31	uw-madison/nystromformer-512	104	BookCorpus and English Wikipedia
32	facebook/opt-350m	316	BookCorpus, CC-Stories, The Pile, Pushshift.io, and CCNewsV2
33	bert-base-uncased	105	BookCorpus and English Wikipedia
34	google/rembert	550	Multilingual Wikipedia data over 110 languages.
35	roberta-base	119	BookCorpus, English Wikipedia, CC-News, OpenWebText, and Stories
36	andreasmadsen/efficient_mlm_m0.40	339	NA
37	ArthurZ/dummy-rochert-qa	112	NA
38	tau/splinter-base	103	BookCorpus and English Wikipedia
39	squeezebert/squeezebert-uncased	49	BookCorpus and English Wikipedia
40	t5-small	58	Colossal Clean Crawled Corpus (C4)
41	xlm-mlm-en-2048	637	Masked language modeling
42	xlnet-base-cased	112	XLNet model pre-trained on English language
43	uw-madison/yoso-4096	121	Masked language modeling
44	SRDdev/QABERT-small	64	30k samples from the Stanford Question Answering Dataset
45	bert-large-uncased-whole-word-masking-finetuned-squad	320	BookCorpus and English Wikipedia
46	facebook/bart-large-cnn	388	CNN Daily Mail
47	ahotrod/electra_large_discriminator_squad2_512	319	SQuAD v2

1

Raster diff (red=missing, blue=extra)



1

Findings and next steps

Which upgrades pay

1-turn -> harness is the big one (+~20 overall; page counts fixed everywhere). Visual feedback and v2 tooling each add several points; extra effort mainly buys reliability, not peak score. Model scale alone changes almost nothing at 1 turn.

v2 tooling

Checkpointing makes bad experiments free; the deterministic sweep found margins/spacing the LLM had hand-nudged for 15 turns - runs that used it posted the two best scores of the experiment.

Where it saturates

Raster sits at its font floor (~20-30); late iterations oscillate by ~1 point. 05_tables_simple_021 resists all variants so far.

Next

v2.1 prompt: always sweep before manual nudges, keep iterating while under budget. Repeat runs for noise; human-calibrate metric weights; then RL hill-climbing with the same deterministic reward.